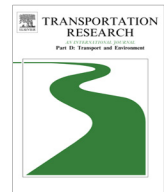




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Environmental efficiency assessment of U.S. transport sector: A slack-based data envelopment analysis approach

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ABSTRACT

Sustainable transportation in the U.S. is essential for long-term economic growth and mobility, and environmental preservation. Using a non-radial slack-based measurement data envelopment analysis (SBM-DEA) model and state-level data, this study assesses the environmental efficiency of the transportation sector in the U.S. from years 2004 to 2012. In addition to environmental efficiency, carbon efficiency and potential carbon reduction were estimated for the 50 U.S. states. The findings of this study reveal that U.S. transportation sector was environmentally inefficient; U.S. states had an average transportation environmental efficiency score below 0.64. Therefore the states could substantially reduce carbon emissions to improve the environmental efficiency of their transportation sectors.

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1. Introduction

Transportation has great influence on the economy of the United States (U.S.). However, one of the most serious issues arising from transportation and economic growth is the environmental impacts across the country, especially increased transportation carbon emissions (Chang, 2013). In recent years, there has been increasing global interest in environmentally sustainability issues. “Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs” (WCED, 1987, Chapter 2, Section IV). Transportation consumes a high amount of energy (Zhou et al., 2014), and sustainable transportation hinges on the ability to maximize transportation environmental performance and to minimize the associated adverse impacts (Hendrickson et al., 2006).

The transportation sector accounted for approximately 10% of the U.S. Gross Domestic Product (GDP) in 2014 (RITA, 2014). The same sector was found to be the second largest source of greenhouse gas (GHG) emissions accounting for 27% of total U.S. GHG emissions, following the power generation industry (US EPA, 2014). An additional critical environmental concern is that energy consumption by the transportation sector is expected to increase dramatically in the next quarter century (Frey and Kuo, 2007). In this context, President Barack Obama initiated a climate action plan that seeks to reduce 17% of total carbon dioxide (CO₂) by 2020 (Leggett, 2014).

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Increasing concerns over the recent environmental issues related to transportation activities have led to sustainable development initiatives becoming a central element of public policy (Egilmez and Park, 2014), and transporting goods and services in a more sustainable way has become an essential topic of discussion. These discussions and projects are expected to contribute to the overall objective of sustainable development (Benjaafar and Savelsbergh, 2014; Choi et al., 2015). Therefore, it is essential to study the environmental performance of transportation activities from a holistic viewpoint to facilitate sustainable development in the transportation sector of a country or a region (Goldman and Gorham, 2006). Recognizing the importance of reducing GHG emissions and energy consumption, a number of studies have evaluated the environmental efficiency of U.S. industries, but they have focused on the environmental efficiency of the industrial sector (Egilmez et al., 2013), freight transportation from a manufacturing perspective (Egilmez and Park, 2014; Park et al., 2015), cross-country comparison (Zhou et al., 2006; Simsek, 2014), and the electricity sector (Barba-Gutiérrez, 2009). No study in the literature has been conducted on the overall environmental performance of the U.S.'s transportation sector.

Given this context, the main objective of this study is to analyze U.S. transportation environmental efficiency over a 9-year period (2004–2012) using a slack-based non-radial data envelopment analysis (SBM-DEA), and to estimate the potential reduction of transportation CO₂ emission. We first measure the environmental efficiency of the transportation sectors in all 50 U.S. states through the SBM-DEA model by incorporating CO₂ as an undesirable output (Chang et al., 2013). More specifically, we estimate carbon efficiency (CE), potential carbon reduction (PCR), excess of inputs and shortfall of output of the U.S. transportation sector. The paper is organized as follows: Section 2 reviews the literature; Section 3 provides the methodology of this study and data description; Section 4 presents the results of the analysis and discussion. Finally, Section 5 provides the conclusion, a discussion on policy implications, and suggests the direction for future research.

2. Literature

Various approaches for measuring environmental efficiency have been proposed in the literature. First of all, one can consider the presence of undesirable outputs using an index number approach. For example, Pittman (1983) extended the study by Caves et al. (1982) incorporating undesirable outputs into a multilateral productivity index. The drawback of this method is the difficulty of measuring the shadow price of undesirable outputs for the productivity index (Chang, 2013; Zhou et al., 2007). Another widely used approach is Data Envelopment Analysis (DEA). DEA has become one of the most used approaches in measuring environmental efficiency due to its robustness in finding optimal efficiency scores for different problems and datasets (Chang, 2013).

As the primary approach, Charnes et al. (1978) proposed the constant returns to scale DEA (CCR-DEA). DEA is a non-parametric approach for estimating the relative efficiency of decision making units (DMUs) by comparing multiple inputs with outputs in the framework of frontier analysis (Cooper et al., 2007). Banker et al. (1984) extended the basic CCR-DEA model to variable returns to scale DEA (BCC-DEA). Since then, DEA has been a popular benchmarking approach commonly used to identify best management practice within a set of DMUs.

In an output-oriented DEA model, an inefficient DMU could expand all its outputs simultaneously and equal-proportionally without increasing its input use. While in an input-oriented model, an inefficient DMU could reduce all its inputs simultaneously and equal-proportionally without sacrificing or reducing its outputs. Hence, the conventional DEA models provide a radial efficiency measure that is either output- or input-oriented (Charnes et al., 1992, 1996; Cook and Seiford, 2009). However, when an environmental pollutant is present in the model, the efficiency assessment becomes a challenging task (Chang, 2013), especially since an environmental pollutant need not increase or decrease equal-proportionally with outputs or inputs (Cooper et al., 2007).

Various methods for modeling undesirable outputs in DEA have been proposed in the literature. One treatment is to consider the pollutant as a free disposable input variable (Hailu and Veeman, 2001), but this concept was challenged by Färe and Grosskopf (2003) who viewed undesirable byproducts as weakly disposable outputs. The concepts of weak disposability and strong disposability of undesirable outputs were proposed by Färe et al. (1989). Under the weak disposability property, a reduction in undesirable outputs will result in a reduction of desirable outputs, while strong disposability also means free disposability, and it assumes that it is possible to reduce the desirable output without reducing the undesirable outputs (Färe et al., 1989). Another approach involves the use of translated input and output data in the traditional BCC-DEA model, and the resulting efficiency classifications remain invariant to data transformation (Seiford and Zhu, 2002; Rousseau and Semple, 1995). In addition to the abovementioned radial approaches, a non-radial DEA model can be used to handle undesirable outputs (Zhou et al., 2007), and a slack-based measurement model proposed by Tone (2001) has also been used to account for the presence of undesirable outputs (Hu and Wang, 2006; Lozano and Gutiérrez, 2011; Chang, 2013; Li et al., 2013).

The slack-based measure (SBM) of efficiency was first proposed by Tone (2001). One main advantage of SBM over the aforementioned radial DEA models is that SBM captures input excesses and output shortfalls of the DMUs while conventional CCR-DEA and BCC-DEA models deal with a proportional reduction or expansion of inputs and outputs (Chang, 2013). Based on the principle of a non-radial model, the primary purpose of the SBM is to locate the DMUs on the efficient frontier, and the objective function of the SBM is to be minimized by finding the maximum slacks (Tone, 2001). Non-radial efficiency SBM-DEA is found to be more appropriate compared to traditional DEA models when it comes to modeling

undesirable byproducts (Zhou et al., 2006; Hernandez-Sancho, 2011) since the traditional models and radial approaches neglect the presence of slacks (Cooper et al., 2007).¹ Zhou et al. (2006) found that SBM-DEA has a higher discriminatory power when compared to the conventional radial efficiency measures, since the radial approach tends to yield a large number of efficient firms with an efficiency score of 1. Another advantage of non-radial SBM-DEA is that the efficiency indicator for each variable can be identified to increase the efficiency level of the DMU being studied (Zhang and Kim, 2014).

The non-radial efficiency SBM-DEA model was applied by Zhang et al. (2008) to the industrial systems in China. The authors measured industrial eco-efficiency by considering the pollutants chemicals' oxygen demand, nitrogen, soot, dust and solid waste as inputs and value added of industries as a desirable output. Besides the pollutants, material and energy consumption were incorporated as inputs in the model as well.

Chang et al. (2013) applied a non-radial efficiency SBM-DEA model to measure the environmental efficiency of the transportation sector in China. They used CO₂ emission as an undesirable output. This approach provided more comprehensive efficiency measures by estimating the economic and environmental performances through capturing the slack values of input and undesirable output as well as the shortfalls of desirable output. Another recent study by Zhou et al. (2014) performed an energy efficiency assessment of the regional transport sectors in China from 2003 to 2009. Some other studies associated with transportation such as a passenger airlines (Merkert and Hensher, 2011), airports (Lin and Hong, 2006), global airlines (Scheraga, 2004), and ports (Chang, 2013; Liu and Lim, 2015) are found in the literature. The only study using the DEA model to assess eco-efficiency of U.S. transportation was conducted by Egilmez and Park (2014). The authors only considered the environmental and economic impacts of transportation from a manufacturing perspective, and environmental impact was incorporated as an input while economic outputs were considered as desirable outputs for assessing eco-efficiency.

The literature shows that environmental impacts are considered as inputs or outputs depending on the type of models used. Based on the literature review, no paper has adopted SBM-DEA approach on measuring the environmental performance of the U.S. transportation sector. There is only a handful of works available in the literature that use SBM-DEA for assessing environmental efficiency, these include an environmental efficiency assessment of OECD countries (Zhou et al., 2006) and a paper on environmental efficiency of transportation activities in China and Korean ports (Chang, 2013). In this study, we apply the SBM-DEA model with a non-radial approach to analyze the environmental efficiency of the transportation sector in U.S. states.

3. Methodology

3.1. Slack-based measure model description

The aim of this study is to develop a framework to measure the environmental efficiency and potential CO₂ reduction of the transportation sector in the U.S. Following Zhou et al. (2006) and Chang (2013), this paper presents a DEA framework based on the slack-based measure (SBM) by incorporating the undesirable output into the objective function and the constraint function (Tone, 2001). We assume that reducing input resources relative to producing more outputs is a criterion for efficiency measurement.

When considering an undesirable output in the model, it should be noted that efficiency can be formed with more desirable output relative to less undesirable output and less input resources (Chang, 2013; Zhang et al., 2011). Suppose that there are $j = \{1, \dots, n\}$ DMUs and that each j uses m inputs to produce p_1 desirable outputs and generate p_2 undesirable outputs (CO₂ emissions). The vectors of inputs, desirable outputs and undesirable outputs for DMU _{i} are given by $x_j \in \mathbb{R}^m$, $y_j \in \mathbb{R}^{p_1}$ and $c_j \in \mathbb{R}^{p_2}$, respectively.

Thus, for n DMU's, we define the input, desirable output and undesirable output matrices as $X = [x_1, \dots, x_n] \in \mathbb{R}^{m \times n}$, Y as $Y = [y_1, \dots, y_n] \in \mathbb{R}^{p_1 \times n}$, C as $C = [c_1, \dots, c_n] \in \mathbb{R}^{p_2 \times n}$. All data on X , Y and C are positive. The production possibility set (PPS) can be described as follows:

$$P(x) = \{(x, y, c) | x \text{ can produce } (y, c), x \geq X\lambda, y \leq Y\lambda, c \geq C\lambda, \lambda \geq 0\}, \quad (1)$$

where λ denotes the non-negative intensity vector, and the production technology in (1) exhibits constant returns to scale (CRS). From the concept of slacks, the efficiency of DMUs must be measured with consideration of how much input waste can be reduced to a given level of output, and how much output can be increased for a given level of input (Tone, 2001). But this original approach developed by Tone (2001) did not consider the presence of any undesirable output in the model. Therefore this study uses a SBM specification that incorporates an undesirable output into both the objective function and an additional constraint on the undesirable output. The SBM-DEA model can thus be expressed in Model 1 below:

¹ A slacks-based directional distance function is a viable alternative to SBM since the directional distance function has an additive structure like the SBM, and it has been shown by Fukuyama and Weber (2009) and Färe and Grosskopf (2003) that the two measures are very much related. Fukuyama and Weber's slacks-based directional distance function (2009) generalized Tone's SBM by normalizing the slacks in inputs and outputs in percentage terms. Färe and Grosskopf (2003) showed that SBM is a special case of their slacks-based directional distance function (DDM), and that "the two efficiency measures have the same indication of efficiency (page 321)." They showed that a DMU is SBM efficient if and only if it is DDM efficient.

$$e_0^* = \frac{1 - \frac{1}{m} \sum_{i=1}^m \frac{s_i^-}{x_{i0}}}{1 + \frac{1}{p_1 + p_2} \left(\sum_{r_1=1}^{p_1} \frac{s_{r_1 0}^y}{y_{r_1 0}} + \sum_{r_2=1}^{p_2} \frac{s_{r_2 0}^c}{c_{r_2 0}} \right)} \quad (2)$$

s.t.

$$x_0 = X\lambda + s^- \quad (3)$$

$$y_0 = Y\lambda - s^y \quad (4)$$

$$c_0 = C\lambda + s^c \quad (5)$$

$$s^- \geq 0, s^y \geq 0, s^c \geq 0, \lambda \geq 0, \quad (6)$$

where,

i = Index of inputs (1, 2, ..., m);

m = Number of inputs;

Subscript '0' = The DMU, whose efficiency is being estimated in the current model;

r_1 = Index of good outputs;

r_2 = Index of bad outputs;

p_1 = Number of good outputs;

p_2 = Number of bad outputs;

s^- = Slack of inputs;

s^y = Slack of good outputs;

s^c = Slack of bad outputs;

The DMU is efficient if e_0^* is equal to 1, which implies all the slack values, s^- , s^y and s^c are equal to 0. If e_0^* is less than 1, then the DMU is inefficient, and it can become efficient by eliminating the slacks in inputs and bad outputs and augmenting the shortfalls in good outputs. But Eq. (2) is not a linear function. Therefore, Model 2, a transformed model which is an equivalent linear programming (LP) model is used (Tone, 2001):

$$r_0^* = \min t - \frac{1}{m} \sum_{i=1}^m \frac{s_i^-}{x_{i0}} \quad (7)$$

s.t.

$$1 = t + \frac{1}{p_1 + p_2} \left(\sum_{r_1=1}^{p_1} \frac{s_{r_1}^y}{y_{r_1 0}} + \sum_{r_2=1}^{p_2} \frac{s_{r_2}^c}{c_{r_2 0}} \right) \quad (8)$$

$$x_0 t = X\beta + s^- \quad (9)$$

$$y_0 t = Y\beta - s^y \quad (10)$$

$$c_0 t = C\beta + s^c \quad (11)$$

$$s^- \geq 0, s^y \geq 0, s^c \geq 0, \beta \geq 0, t > 0, \quad (12)$$

The optimal solution of the LP model (7–12) can be solved, and let the optimal solution be (r^* , t^* , β^* , s^{-*} , s^{y*} , s^{c*}) where $e_0^* = r_0^*$, $\lambda^* = \frac{\beta^*}{t^*}$, $s^{-*} = \frac{s_i^-}{t^*}$, $s^{y*} = \frac{s_{r_1}^y}{t^*}$, $s^{c*} = \frac{s_{r_2}^c}{t^*}$ from Model (2). The solution of t^* , β^* , s^{-*} , s^c and s^y can be generated through Model 2 with $t^* > 0$.

In this paper, carbon efficiency (CE) of each DMU is estimated based on the method proposed by Hu and Wang (2006) where the index of total factor energy efficiency was introduced using DEA-generated optimal energy input level, and by Zhou and Ang's approach (2008) of evaluating energy efficiency with undesirable output. The carbon efficiency (CE) can be estimated as follows (Chang et al., 2013):

$$CE = \text{Target carbon emission} / \text{Real carbon emission} = \frac{C_0^t - S_0^c}{C_0^t}, \quad (13)$$

where C_0^t is the observed carbon emission, and S_0^c is the slack of carbon emission, therefore $C_0^t - S_0^c$ is the target carbon emission level. Additionally, the potential carbon reduction (PCR) of each state is estimated by the slack variable S_0^c as it is the excess carbon emission. Finally, the performance improvement of each input and output indicator is evaluated in percentage terms.

3.2. Data description

In many empirical studies, capital, energy and labor are considered three major inputs in production, and gross domestic product (GDP) is a measure of the overall economic output of a sector or an economy. In order to analyze the environmental efficiency of the U.S. transportation sector, this study uses a panel data set of all 50 U.S. states from 2004 to 2012. The data include capital expense, energy consumption and labor in the transportation sector as input variables. The labor and capital

input data were collected from the U.S. Bureau of Labor Statistics and the U.S. Census Bureau. The data on the volume of energy consumed by the transportation sector were collected from the U.S. Energy Information Administration. Each state's transportation value added (GDP) was considered a desirable output (Chang, 2013; Zhou et al., 2014), and the data were collected from the Bureau of Economic Analysis. Additionally, an undesirable output, CO₂, is also taken into account as a byproduct of producing transportation services (Simsek, 2014). The data on CO₂ emissions were available from the U.S. Energy Information Administration (U.S. EIA). The data descriptions are provided in Table 1.

4. Results and discussion

4.1. Input and output indicators

Table 2 shows the descriptive statistics of the state-level data from 2004 to 2012. The capital expenditure of U.S. states' transportation sectors averaged 4.76 million dollars for 2004–2012. The average state transportation sector consumed 599.5 trillion Btu of energy, employed 168 thousand people, produced 8090 million dollars in transportation GDP (value-added) and emitted 37.9 million metric tons of CO₂. Variations in the input and output variables across the states can be seen from the standard deviations in Table 2. The correlation matrix of inputs and outputs in Table 3 is analyzed to see if there is a significant relationship between the input and output variables. From the results in Table 3, we can see that a significantly high correlation exists between the input and the output variables in that the correlation coefficients are all above 0.60.

4.2. U.S. transportation environmental efficiency performance

As mentioned in Section 3, the environmental efficiency (EE) score in the transportation sector is evaluated by e_0^* in Eq. (2), because it includes the slack values of all input and output variables. The carbon efficiency (CE) score is estimated by Eq. (13), and finally potential carbon reduction (PCR) is calculated by the slack variable s_0^* . Tables 4 and 5 show the results of EE and CE indicators for each U.S. state from 2004 to 2012.

The overall average EE performance from 2004 to 2012 of the transportation sector in the U.S. indicates that only four of the fifty states (Alaska, Illinois, Nebraska and Vermont) were found to be relatively environmentally efficient as scores of EE in the four states are ranked as the best in the country (EE = 1). The EE scores of the inefficient states ranged from 0.341 to 0.965 (average = 0.640), with Texas ranking first and Alabama ranking last among the inefficient states. The EE scores suggest considerable room for transportation environmental efficiency improvements in most states.

Table 1
Input and output variables and data sources, 2004–2012.

	Variables	Unit	Sources
Input	Capital expenses	In millions	U.S. Census Bureau
	Energy use	In trillion Btu	U.S. Energy Information Administration
	Labor	In thousands (person)	U.S. Bureau of Labor Statistics
Output	Desirable output: Value added (GDP)	In millions of U.S. dollars	Bureau of Economic Analysis
	Undesirable output: CO ₂ emission	In millions of metric tons	U.S. Energy Information Administration

Table 2
Descriptive statistics of input and output, 2004–2012.

Variable	Unit	Min	Max	Mean	Std. Dev.
Capital	10 ⁶ dollar	3.8	30,312.6	4,762.6	5,301.7
Energy	10 ¹² btu	19.6	3,387.3	599.5	697.8
Labor	10 ³ persons	7	951	167.7	178.6
GDP (value-added)	10 ⁶ dollar	298	53,443.0	8,090.0	8,982.6
CO ₂	10 ⁶ ton	1.1	238.1	37.9	41.6

Table 3
Correlation matrix of inputs and outputs.

	Capital	Energy	Labor	CO2	GDP
Capital	1.00				
Energy	0.67	1.00			
Labor	0.84	0.80	1.00		
CO2	0.81	0.83	0.96	1.00	
GDP	0.83	0.81	0.97	0.96	1.00

Table 4

Environmental efficiency based on SBM, 2004–2012.

State	2004	2005	2006	2007	2008	2009	2010	2011	2012	Mean
Alaska	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Illinois	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Nebraska	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Vermont	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Texas	1.000	1.000	1.000	0.686	1.000	1.000	1.000	1.000	1.000	0.965
Wyoming	0.838	0.984	1.000	0.818	1.000	1.000	1.000	1.000	1.000	0.960
California	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.825	0.776	0.956
Hawaii	1.000	1.000	1.000	0.758	0.748	0.820	0.911	0.842	1.000	0.898
Rhode Island	0.797	0.841	0.851	0.868	0.844	1.000	0.873	1.000	1.000	0.897
New Jersey	0.832	0.861	0.779	1.000	1.000	1.000	0.832	0.771	0.790	0.874
Tennessee	1.000	1.000	1.000	0.302	1.000	0.896	0.720	0.854	1.000	0.864
Delaware	0.747	0.734	0.789	1.000	0.719	0.764	1.000	0.879	0.834	0.830
Georgia	0.869	0.774	0.677	0.708	0.846	0.890	0.676	1.000	1.000	0.827
North Dakota	0.737	0.813	1.000	0.713	0.812	0.917	0.811	0.849	0.745	0.822
New York	0.739	0.791	0.726	0.666	0.820	0.831	1.000	0.813	0.799	0.798
Montana	0.683	0.728	0.709	0.644	0.672	0.740	0.747	0.742	0.712	0.709
South Dakota	0.700	0.711	0.746	0.659	0.672	0.684	0.700	0.662	0.626	0.684
Ohio	0.650	0.713	0.652	0.527	0.742	0.742	0.781	0.661	0.662	0.681
Pennsylvania	0.760	0.703	0.627	0.548	0.699	0.695	0.683	0.656	0.650	0.669
Florida	0.620	0.653	0.600	0.554	0.708	0.747	0.613	0.706	0.669	0.652
New Hampshire	0.645	0.715	0.680	0.580	0.588	0.618	0.595	0.674	0.695	0.643
Idaho	0.574	0.596	0.596	0.593	0.547	0.577	0.691	0.579	0.602	0.595
Maine	0.529	0.538	0.567	0.519	0.577	0.605	0.743	0.598	0.595	0.586
Virginia	0.520	0.504	0.535	1.000	0.500	0.585	0.514	0.526	0.534	0.580
Indiana	0.582	0.609	0.595	0.623	0.553	0.561	0.513	0.550	0.569	0.573
Nevada	0.565	0.568	0.531	0.501	0.554	0.597	0.553	0.642	0.631	0.571
Connecticut	0.506	0.535	0.530	0.619	0.571	0.580	0.588	0.592	0.579	0.567
Arkansas	0.538	0.572	0.580	0.624	0.519	0.525	0.569	0.535	0.561	0.558
Washington	0.575	0.591	0.547	0.465	0.516	0.557	0.569	0.578	0.587	0.554
Kentucky	0.583	0.672	0.544	0.677	0.436	0.509	0.504	0.500	0.516	0.549
New Mexico	0.475	0.453	0.444	0.410	0.467	0.482	1.000	0.569	0.547	0.539
Kansas	0.502	0.526	0.517	0.601	0.505	0.520	0.558	0.548	0.569	0.538
Missouri	0.568	0.589	0.555	0.459	0.558	0.550	0.545	0.468	0.505	0.533
Utah	0.565	0.602	0.549	0.528	0.492	0.513	0.497	0.499	0.492	0.526
Louisiana	0.454	0.432	0.469	0.603	0.553	0.569	0.507	0.470	0.541	0.511
Michigan	0.568	0.549	0.496	0.432	0.486	0.463	0.453	0.526	0.553	0.503
Minnesota	0.582	0.567	0.467	0.591	0.438	0.438	0.424	0.461	0.461	0.492
West Virginia	0.458	0.462	0.480	0.431	0.478	0.486	0.558	0.517	0.495	0.485
Arizona	0.459	0.516	0.466	0.581	0.367	0.410	0.469	0.483	0.503	0.473
Iowa	0.487	0.464	0.459	0.542	0.441	0.451	0.471	0.452	0.453	0.469
Massachusetts	0.405	0.404	0.388	1.000	0.367	0.403	0.405	0.412	0.429	0.468
Wisconsin	0.540	0.540	0.493	0.449	0.395	0.436	0.422	0.415	0.427	0.457
North Carolina	0.479	0.487	0.454	0.391	0.429	0.483	0.383	0.430	0.466	0.445
Oregon	0.491	0.472	0.449	0.417	0.423	0.421	0.357	0.445	0.474	0.439
Oklahoma	0.454	0.452	0.436	0.353	0.353	0.368	0.596	0.408	0.410	0.426
Maryland	0.427	0.416	0.386	0.484	0.356	0.375	0.408	0.396	0.365	0.402
Mississippi	0.359	0.399	0.349	0.339	0.334	0.387	0.405	0.373	0.367	0.368
Colorado	0.347	0.396	0.368	0.418	0.340	0.351	0.337	0.342	0.389	0.365
South Carolina	0.317	0.337	0.343	0.434	0.342	0.375	0.364	0.337	0.347	0.355
Alabama	0.340	0.338	0.336	0.357	0.329	0.329	0.354	0.337	0.344	0.341
Mean	0.637	0.652	0.635	0.629	0.622	0.645	0.654	0.638	0.645	0.640

In terms of CE, five states (the four previously mentioned states and Texas) were found to be (relatively) carbon efficient states. CE scores for the inefficient states ranged from 0.307 to 0.975 (average = 0.638) with Rhode Island ranking first and South Carolina ranking last among the inefficient states. In Figs. 1 and 2, we use graduated colors with equal interval labels to present the annual average EE and CE scores of the states during the study period. It can be easily observed that there is a consistent spatial distribution pattern of EE and CE, in that states that are low on EE are also low on CE.

Taking all 50 states' performance as a whole, Fig. 3 displays the sector's mean EE and CE scores by year. The EE and CE performance fluctuated every year. The average EE scores hovered between 0.62 and 0.66. There was a decreasing trend in EE from 2004 to 2008, and the average percentage change was found to be -0.58% . However, after 2008, the transportation sector exhibited an upward trend in EE. The sector's CE performance slid to 0.6 in 2008, but rose to nearly 0.69 in the next 2 years. The sector's EE performance also was lowest in 2008 but rose above 0.65 in 2010. The EE trend reflects the rate of CO₂ emissions by the U.S. transportation sector during the same period (US EPA, 2014).

Table 5

Carbon efficiency based on SBM, 2004–2012.

State	2004	2005	2006	2007	2008	2009	2010	2011	2012	Mean
Alaska	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Nebraska	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Illinois	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Vermont	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Texas	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Rhode Island	0.942	0.955	0.950	1.000	0.932	1.000	1.000	1.000	1.000	0.975
California	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.913	0.856	0.974
Wyoming	0.767	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.972
Tennessee	1.000	1.000	1.000	0.735	1.000	0.871	0.777	0.870	1.000	0.917
New York	0.852	0.967	0.838	0.929	0.895	0.924	1.000	0.931	0.872	0.912
Delaware	0.860	0.822	0.832	1.000	0.787	0.831	1.000	0.915	0.903	0.883
Hawaii	1.000	1.000	1.000	0.916	0.665	0.705	0.929	0.724	1.000	0.882
New Jersey	0.714	0.783	0.619	1.000	1.000	1.000	0.835	0.806	0.833	0.843
North Dakota	0.731	0.781	1.000	0.926	0.754	0.893	0.908	0.765	0.758	0.835
Georgia	0.774	0.772	0.587	0.792	0.716	0.902	0.736	1.000	1.000	0.809
Pennsylvania	0.713	0.769	0.650	0.692	0.703	0.720	0.794	0.704	0.701	0.716
South Dakota	0.734	0.737	0.737	0.842	0.682	0.673	0.741	0.629	0.599	0.708
Montana	0.665	0.638	0.634	0.691	0.623	0.666	0.857	0.671	0.736	0.687
Ohio	0.600	0.774	0.615	0.645	0.643	0.641	0.716	0.582	0.600	0.646
Nevada	0.561	0.550	0.516	0.537	0.544	0.636	0.794	0.791	0.835	0.640
New Hampshire	0.599	0.657	0.668	0.754	0.567	0.584	0.668	0.586	0.611	0.633
Idaho	0.601	0.609	0.580	0.822	0.572	0.588	0.684	0.564	0.587	0.623
Florida	0.562	0.637	0.564	0.614	0.643	0.674	0.687	0.586	0.570	0.615
Indiana	0.512	0.637	0.507	0.726	0.495	0.502	0.543	0.573	0.582	0.564
Maine	0.547	0.510	0.524	0.638	0.550	0.541	0.655	0.533	0.565	0.563
Washington	0.535	0.570	0.510	0.495	0.498	0.536	0.629	0.573	0.565	0.546
Connecticut	0.441	0.463	0.485	0.538	0.530	0.544	0.608	0.607	0.668	0.543
Kansas	0.512	0.537	0.506	0.514	0.497	0.485	0.593	0.567	0.637	0.539
Arkansas	0.493	0.492	0.487	0.821	0.445	0.439	0.547	0.502	0.561	0.532
Utah	0.523	0.527	0.479	0.519	0.482	0.510	0.593	0.528	0.596	0.529
Wisconsin	0.588	0.615	0.543	0.522	0.437	0.471	0.525	0.515	0.498	0.524
West Virginia	0.471	0.467	0.472	0.536	0.512	0.508	0.646	0.530	0.531	0.519
Missouri	0.525	0.610	0.493	0.490	0.478	0.486	0.501	0.455	0.511	0.506
Michigan	0.511	0.621	0.441	0.464	0.426	0.411	0.486	0.560	0.556	0.497
Kentucky	0.488	0.590	0.472	0.453	0.427	0.440	0.510	0.510	0.506	0.488
Minnesota	0.552	0.539	0.442	0.466	0.434	0.446	0.503	0.508	0.482	0.486
Iowa	0.488	0.472	0.465	0.470	0.440	0.465	0.527	0.486	0.554	0.485
Colorado	0.449	0.482	0.400	0.781	0.395	0.416	0.482	0.470	0.479	0.484
New Mexico	0.382	0.393	0.384	0.442	0.409	0.409	1.000	0.443	0.461	0.480
Virginia	0.431	0.482	0.430	0.425	0.422	0.471	0.592	0.491	0.557	0.478
Arizona	0.405	0.442	0.406	0.706	0.372	0.408	0.515	0.496	0.498	0.472
Oregon	0.446	0.440	0.425	0.423	0.403	0.399	0.476	0.467	0.524	0.445
North Carolina	0.454	0.535	0.417	0.404	0.373	0.412	0.510	0.425	0.453	0.442
Massachusetts	0.405	0.382	0.366	0.370	0.340	0.380	0.444	0.432	0.449	0.396
Maryland	0.388	0.371	0.365	0.370	0.351	0.357	0.450	0.437	0.465	0.395
Louisiana	0.280	0.312	0.297	0.367	0.409	0.485	0.490	0.420	0.469	0.392
Oklahoma	0.342	0.320	0.312	0.530	0.293	0.304	0.421	0.382	0.416	0.369
Alabama	0.283	0.285	0.283	0.524	0.282	0.289	0.347	0.316	0.344	0.328
Mississippi	0.303	0.305	0.286	0.310	0.282	0.297	0.383	0.327	0.338	0.315
South Carolina	0.270	0.289	0.275	0.501	0.260	0.256	0.325	0.283	0.307	0.307
Mean	0.614	0.642	0.605	0.674	0.599	0.620	0.689	0.638	0.661	0.638

4.3. Potential carbon emission reduction of transportation sector

As the results of EE and CE indicate, most of the states are not performing efficiently in the transportation sector, leading to the conclusion that there is great potential to reduce transportation CO₂ emissions in each state. We can see in Table 6 that the U.S. transportation sector could reduce a great deal of CO₂ ranging from at least 0.03 million metric tons to 23.40 million metric tons. The average PCR was found to be 7.10 million metric tons. As shown in the last column of Table 6, on average, 46 U.S. states' transportation sectors showed excessive CO₂ emissions that need to be reduced. Among the states, Florida shows the highest potential for carbon reduction with 23.40 million metric tons, followed by Louisiana with 23.18 million metric tons and North Carolina with 20.33 million metric tons. Compared to Louisiana and North Carolina, Florida had a relatively higher EE score but also a large PCR. This suggests that the inefficiency in Florida's transportation sector can be explained in large part by the presence of environmental slack. On the other hand, North Dakota, Tennessee, Delaware and Rhode Island were found to have only a small amount of excess CO₂ emissions, showing 0.62, 0.61, 0.29 and 0.03 million metric tons of PCR, respectively.

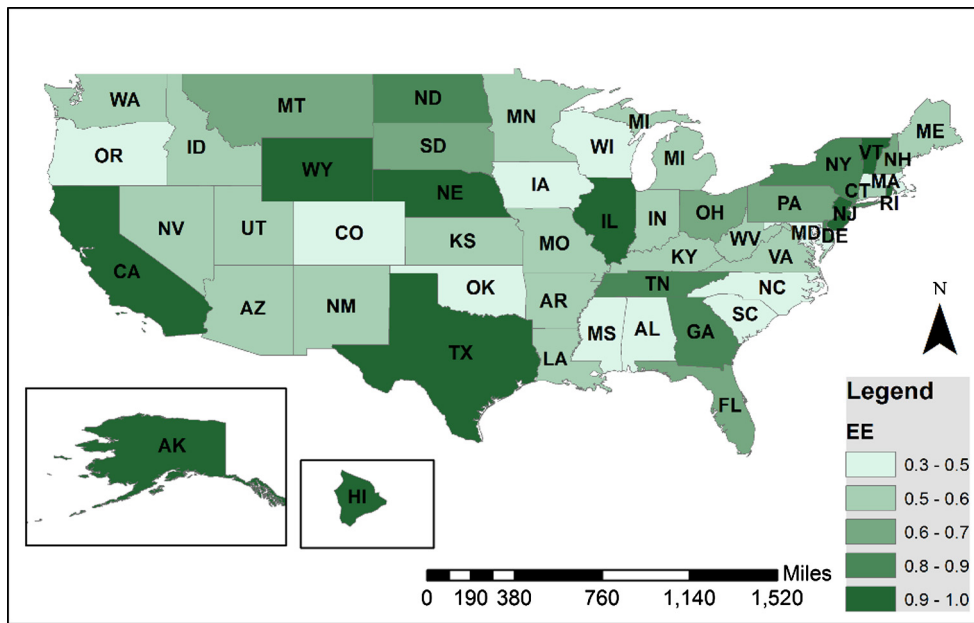


Fig. 1. Spatial distribution of average environmental efficiency values for the transportation sector in 50 U.S. states between 2004 and 2012.

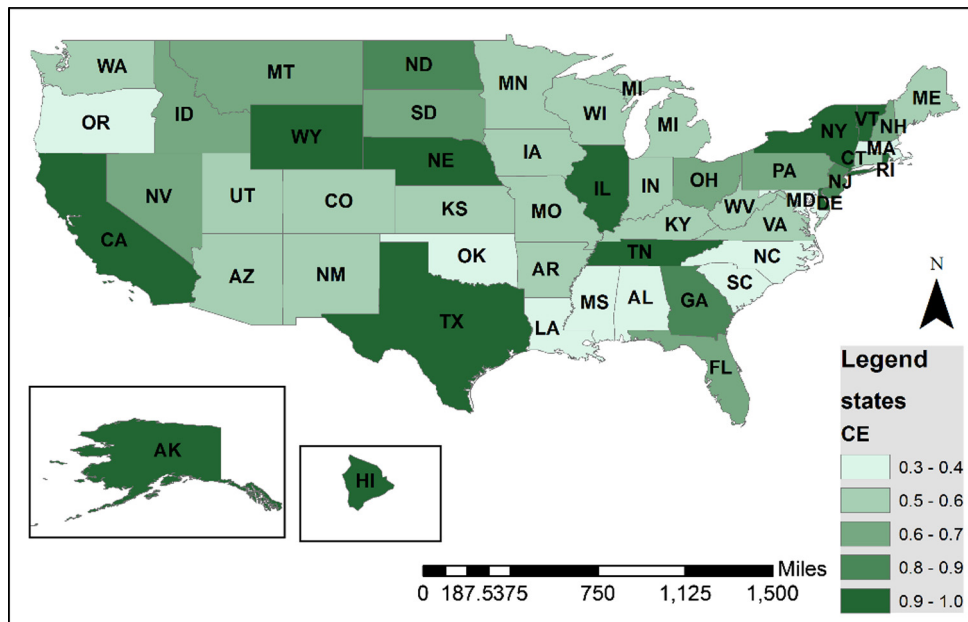


Fig. 2. Spatial distribution of average carbon efficiency values for the transportation sector in 50 U.S. states between 2004 and 2012.

4.4. Analysis of slack variables

There are only four environmentally efficient states, and most EE scores are quite low. Therefore, it is imperative for us to examine the slack values of the inputs and outputs in the model. The purpose of measuring relative efficiency is to determine the amount of excess inputs and the shortfall of output so that inefficient DMUs can identify the best management practices in developing and maintaining a sustainable transportation system. The estimated slack values and the associated percentages of improvements (reported in parentheses) are presented in Table 7. The negative values in the parentheses in Table 7 indicates the percentage of potential input reduction, and the positive values in the parentheses implies the potential increase in percentage terms.

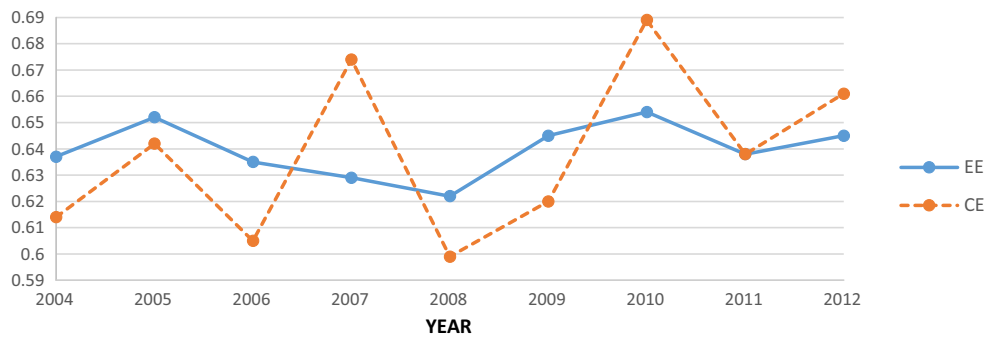


Fig. 3. Average environmental efficiency and carbon efficiency scores of U.S. transportation sector, 2004–2012.

Table 6

Potential carbon reductions, 2004–2012.

State	2004	2005	2006	2007	2008	2009	2010	2011	2012	Mean
Florida	21.53	18.48	48.46	44.11	7.39	4.53	29.60	13.86	22.64	23.40
Louisiana	27.01	24.56	26.27	19.02	23.62	23.13	14.46	27.64	22.90	23.18
North Carolina	17.22	18.94	18.58	14.79	28.59	24.14	17.49	22.72	20.49	20.33
Michigan	16.40	17.32	18.75	12.93	23.26	24.55	15.19	18.67	19.06	18.46
Virginia	19.82	22.21	20.05	15.67	23.99	19.93	7.22	17.12	19.91	18.43
Alabama	16.19	15.99	16.31	8.50	24.11	23.11	11.99	23.01	21.34	17.84
South Carolina	15.98	14.16	15.73	4.79	22.69	23.11	13.78	22.17	19.44	16.87
Oklahoma	10.13	12.13	12.83	7.54	22.86	21.50	5.04	19.74	18.87	14.52
Mississippi	11.52	11.38	13.11	5.83	18.42	17.44	12.39	16.61	14.77	13.50
Massachusetts	8.32	9.33	9.05	0.00	22.07	19.06	8.21	17.58	16.99	12.29
Indiana	10.73	10.58	10.51	4.58	15.78	15.51	8.88	16.14	15.51	12.02
Arizona	10.00	10.26	10.32	5.40	21.58	18.98	0.00	14.79	14.46	11.76
Ohio	12.73	8.52	18.73	16.59	5.89	5.62	16.82	8.66	11.43	11.67
Missouri	8.06	9.17	9.00	3.47	16.53	16.32	5.34	18.53	17.66	11.56
Maryland	7.56	8.78	8.95	1.72	19.90	20.39	0.52	16.48	16.00	11.14
Minnesota	4.73	6.35	7.77	4.42	17.90	16.75	3.22	14.18	15.60	10.10
Washington	8.77	9.07	9.56	7.71	15.65	13.44	0.03	10.20	11.58	9.56
Colorado	4.46	4.40	6.24	1.16	18.25	17.12	0.00	15.32	14.98	9.10
Kentucky	6.11	4.57	5.25	0.74	17.53	17.18	0.02	15.49	14.04	8.99
Oregon	3.39	3.71	4.43	0.56	13.60	13.64	6.04	11.24	9.36	7.33
Wisconsin	1.16	0.92	1.66	0.00	16.46	14.77	2.85	13.51	14.11	7.27
Iowa	1.43	2.14	2.33	0.12	12.07	11.21	2.51	11.15	8.88	5.76
New Mexico	5.90	5.71	6.22	3.51	8.43	8.03	0.00	7.74	6.08	5.73
Arkansas	1.22	1.26	1.37	0.31	11.39	11.30	0.21	10.03	7.76	4.98
Georgia	3.93	9.58	17.90	9.92	0.14	2.02	0.28	0.00	0.00	4.86
Pennsylvania	0.00	1.46	17.15	13.96	0.00	0.00	7.97	0.00	2.87	4.82
California	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.81	25.29	4.79
New Jersey	5.20	8.08	16.64	0.00	0.00	0.00	0.00	5.95	5.81	4.63
Kansas	0.54	0.00	0.75	0.00	9.59	10.08	0.00	8.22	6.41	3.95
Utah	0.40	0.35	2.04	0.00	8.83	7.90	0.36	8.17	5.68	3.75
Connecticut	3.47	2.70	1.90	0.00	7.89	7.34	0.00	6.15	4.20	3.74
West Virginia	3.02	3.35	3.34	0.19	5.39	5.33	0.51	5.16	4.13	3.38
Maine	2.29	3.06	2.80	0.36	3.69	3.64	2.54	3.75	3.14	2.81
Nevada	0.00	0.00	0.52	0.00	7.46	5.27	0.00	2.74	2.04	2.00
Idaho	1.02	1.01	1.49	0.23	3.76	3.30	0.37	3.83	2.91	1.99
New Hampshire	1.64	0.90	0.84	0.08	3.15	2.69	1.55	2.76	2.52	1.79
New York	0.00	0.00	8.80	3.73	0.00	0.00	0.00	0.00	0.00	1.39
South Dakota	0.59	0.55	0.59	0.52	1.93	1.82	1.54	2.26	2.68	1.39
Montana	0.21	0.60	0.64	0.72	3.14	2.62	1.05	2.27	1.07	1.37
Hawaii	0.00	0.00	0.00	0.32	3.25	2.72	0.23	2.82	0.00	1.04
North Dakota	0.30	0.17	0.00	0.53	1.49	0.06	0.64	1.55	0.87	0.62
Tennessee	0.00	0.00	0.00	3.40	0.00	1.41	0.00	0.65	0.00	0.61
Delaware	0.01	0.28	0.19	0.00	1.07	0.54	0.00	0.17	0.35	0.29
Rhode Island	0.00	0.00	0.00	0.00	0.28	0.00	0.00	0.00	0.00	0.03
Texas	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wyoming	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nebraska	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Vermont	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Illinois	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Alaska	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean	5.46	5.64	7.54	4.35	9.78	9.15	8.98	9.14	8.88	7.10

Table 7

Summary of average excess in inputs and shortfall in outputs, 2004–2012.

State	Inputs (Excess)						Undesirable output (excess)		Desirable output (shortfall)	
	Capital (\$)	Slack (%)	Energy (Btu)	Slack (%)	Labor (person)	Slack (%)	CO ₂ (Ton)	Slack (%)	GDP (\$)	Slack (%)
Alabama	960010.8	(−21.3)	1.6	(0.0)	30.5	(−14.4)	17.8	(−52.0)	2107.9	(32.9)
Alaska	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)
Arizona	2002066.5	(−43.2)	9.5	(−1.7)	24.3	(−14.3)	11.8	(−33.8)	209.6	(1.2)
Arkansas	664862.4	(−24.5)	0.3	(−0.1)	22.7	(−19.5)	5.0	(−24.4)	51.9	(0.8)
California	9333921.1	(0.0)	25.3	(0.0)	144.0	(0.0)	4.8	(−2.2)	2134.7	(0.0)
Colorado	1894182.3	(−46.8)	2740.2	(−86.5)	16.4	(−13.3)	9.1	(−30.5)	62.3	(0.5)
Connecticut	995853.6	(−38.7)	6.7	(−3.3)	16.3	(−25.3)	3.7	(−21.9)	63.8	(0.5)
Delaware	668498.0	(−20.9)	0.7	(−0.4)	6.9	(−9.5)	0.3	(−6.0)	86.3	(3.0)
Florida	8996813.9	(−29.2)	6.1	(−0.3)	119.9	(−4.1)	23.4	(−21.6)	1438.4	(3.9)
Georgia	1679037.6	(−11.1)	1.1	(0.0)	73.6	(−3.3)	4.9	(−7.4)	77.3	(0.0)
Hawaii	103764.6	(0.0)	0.0	(0.0)	0.0	(0.0)	1.0	(−9.2)	0.0	(0.0)
Idaho	656505.5	(−34.7)	0.6	(−0.8)	14.2	(−32.1)	2.0	(−22.0)	301.9	(27.7)
Illinois	4432319.3	(0.0)	6.0	(0.0)	110.8	(0.0)	0.0	(0.0)	0.0	(0.0)
Indiana	1393073.0	(−28.4)	4.0	(−0.3)	67.9	(−26.2)	12.0	(−27.5)	315.7	(4.0)
Iowa	1714996.1	(−48.7)	2.0	(−0.7)	45.4	(−40.1)	5.8	(−26.9)	162.9	(6.0)
Kansas	1602683.8	(−50.0)	3.2	(−1.8)	12.5	(−18.6)	4.0	(−20.7)	0.0	(0.4)
Kentucky	1636427.3	(−36.2)	2.2	(−0.3)	30.6	(−15.8)	9.0	(−27.1)	95.4	(2.7)
Louisiana	554242.0	(0.0)	0.2	(0.0)	2.3	(0.0)	23.2	(−46.2)	791.9	(0.0)
Maine	648567.3	(−28.0)	0.5	(−0.5)	11.3	(−22.3)	2.8	(−32.3)	406.1	(50.3)
Maryland	3132041.5	(−56.9)	8.7	(−2.1)	19.9	(−16.4)	11.1	(−36.0)	255.8	(2.3)
Massachusetts	2618724.1	(−52.4)	8.9	(−1.9)	28.5	(−16.9)	12.3	(−38.1)	246.3	(3.2)
Michigan	2491423.2	(−38.4)	15.8	(−1.9)	75.8	(−24.8)	18.5	(−35.2)	534.6	(4.7)
Minnesota	3663932.0	(−59.2)	9.5	(−2.0)	24.6	(−12.3)	10.1	(−29.3)	69.0	(0.4)
Mississippi	1002122.2	(−31.7)	2.2	(−0.5)	10.7	(−13.2)	13.5	(−52.7)	1337.2	(37.5)
Missouri	1949786.2	(−37.2)	2.5	(−0.2)	44.5	(−21.6)	11.6	(−28.4)	456.3	(7.4)
Montana	1145352.6	(−43.4)	0.9	(−0.7)	1.5	(−4.9)	1.4	(−16.5)	79.5	(16.7)
Nebraska	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)
Nevada	4848524.0	(−60.1)	1.3	(−0.8)	14.7	(−23.3)	2.0	(−12.7)	0.0	(0.3)
New Hampshire	920129.6	(−32.3)	0.9	(−0.8)	7.7	(−16.3)	1.8	(−24.6)	188.8	(39.2)
New Jersey	1286622.8	(−8.5)	3.4	(0.0)	33.6	(−0.5)	4.6	(−7.0)	8.7	(0.0)
New Mexico	630779.4	(−25.9)	2.2	(−0.9)	6.3	(−16.5)	5.7	(−38.8)	755.6	(39.2)
New York	10321008.7	(−45.6)	20.4	(−0.5)	175.5	(−20.6)	1.4	(−1.9)	0.0	(0.0)
North Carolina	4344011.3	(−44.0)	5.6	(−0.6)	82.3	(−21.0)	20.3	(−39.8)	1038.6	(9.9)
North Dakota	1249116.5	(−31.2)	3.2	(−2.0)	1.0	(−1.4)	0.6	(−8.9)	95.8	(8.8)
Ohio	2854580.2	(−31.9)	8.1	(−0.5)	138.8	(−26.5)	11.7	(−17.0)	62.4	(2.1)
Oklahoma	761736.8	(−22.9)	3.5	(−0.8)	5.4	(−8.7)	14.5	(−46.1)	559.4	(6.8)
Oregon	2970890.4	(−50.9)	2.1	(−0.6)	24.5	(−23.3)	7.3	(−32.2)	141.9	(2.4)
Pennsylvania	6901247.5	(−54.7)	9.8	(−0.4)	159.6	(−27.2)	4.8	(−7.0)	0.0	(1.0)
Rhode Island	765612.9	(−17.3)	1.9	(−0.3)	11.5	(−20.2)	0.0	(−0.7)	68.1	(0.0)
South Carolina	813063.1	(−19.7)	1.1	(−0.1)	23.2	(−6.9)	16.9	(−53.9)	1602.3	(42.8)
South Dakota	1128334.9	(−47.3)	1.4	(−1.4)	2.5	(−5.0)	1.4	(−21.9)	179.0	(34.0)
Tennessee	943627.2	(−19.0)	1.1	(−0.2)	18.3	(−1.5)	0.6	(−1.4)	0.0	(0.0)
Texas	5494902.7	(0.0)	8.8	(0.0)	74.1	(0.0)	0.0	(0.0)	882.7	(0.0)
Utah	1542717.3	(−48.8)	0.9	(−0.6)	18.8	(−24.7)	3.7	(−21.8)	106.7	(3.2)
Vermont	878603.2	(0.0)	0.4	(0.0)	4.4	(0.0)	0.0	(0.0)	149.8	(0.0)
Virginia	2891505.3	(−43.3)	6.8	(−0.7)	24.4	(−9.1)	18.4	(−34.7)	758.3	(3.3)
Washington	4156807.1	(−55.9)	1.7	(−0.1)	43.4	(−11.0)	9.6	(−22.1)	168.1	(3.3)
West Virginia	1274187.0	(−52.8)	4.9	(−3.3)	15.4	(−32.8)	3.4	(−28.4)	251.7	(29.2)
Wisconsin	3656673.2	(−62.2)	3.0	(−1.1)	78.0	(−37.0)	7.3	(−24.1)	0.0	(0.0)
Wyoming	329295.6	(−11.3)	2.1	(−0.2)	1.1	(0.0)	0.0	(0.0)	0.0	(0.0)
Mean	2338103.7	(−31.3)	59.1	(−2.4)	38.4	(−14.0)	7.1	(−21.9)	366.1	(8.6)

Note: Slack (%) = (Target – Actual)/Actual × 100.

By comparing the results in Tables 4–7, we can see that low-ranked environmentally inefficient states have extremely high input slacks in both level and percentage terms. This pattern suggests that in order to increase efficiency the input levels need to be lowered by the estimated input slacks in Table 7. Meanwhile, carbon emissions excess needs to be cut, and good output shortfall should also be eliminated. For example, among the environmentally inefficient states, Alabama has excesses in capital, labor and CO₂, while producing insufficient transportation output. The state could increase its transportation output by 33% per year, while reducing its carbon emissions by as much as 52% per year, and slashing its capital and labor inputs by 21% and 14%, respectively. Other inefficient states, such as Maryland, Mississippi, Colorado and South Carolina also show much waste in input variables and high shortfalls in transportation GDP. In addition, the third-best state among the inefficient states, California, has extremely high excess values in capital, energy and labor, as well as a shortfall in transportation GDP. Florida and Louisiana show the greatest excess in undesirable output (CO₂). On average, capital investment has the

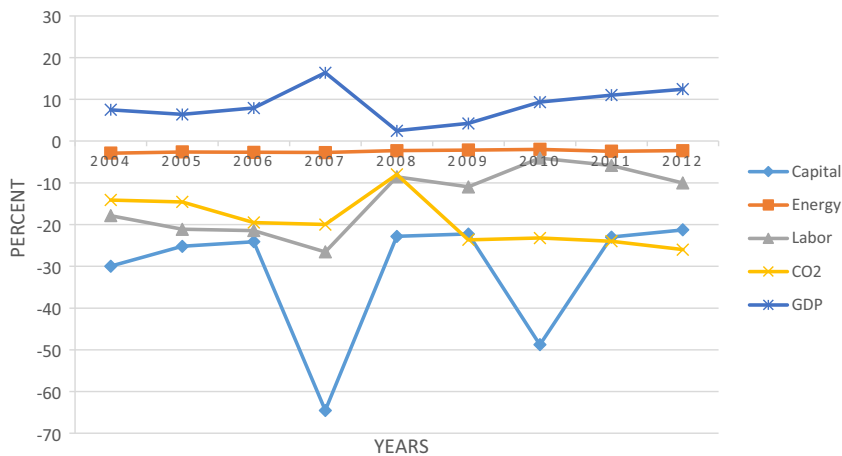


Fig. 4. The percentage change of input excesses, and output shortfall.

highest percentage of slack at -31.3% , CO_2 has the second highest in slack at -21.9% , and labor has the third highest in slack at -14% , while energy input has relatively less slack at -2.4% .

Fig. 4 represents the dynamic changes of slacks in inputs, desirable output, and undesirable output from 2004 to 2012. A positive percentage change indicates the shortfall of desirable output, and a negative percentage change indicates the excesses in inputs and undesirable output that need to be reduced. The average shortfall in transportation GDP peaked at 16% in 2007 and dropped to a minimum in 2008, after which it continually increased until 2012. Additionally, the percentages of average excesses in capital and labor fluctuated every year. The large capital excess in 2007 coupled with the output shortfall in the same year suggest that the industry may have overinvested during a time when the overall U.S. economic growth was slowing down. As the U.S. economy showed signs of recovery, and as a result of a direct fiscal stimulus in 2009 (Federal Reserve Bank of St. Louis, Undated; Transportation Research Board, 2014), excess capital spiked in 2010. At the same time, the CO_2 slack largely mirrored the movement of transportation GDP and reached a minimum in 2008. Other than 2008, the sector's CO_2 slack was prevalent over the study period. This result indicates that U.S. policy on transportation carbon emissions has yet to improve. The slack in energy is negligible, which shows little dynamic change over the study period.

5. Summary and conclusion

Sustainable development in U.S. transportation is not only essential for economic growth and mobility, but also for the environment. However, no previous studies have been conducted on the environmental efficiency of the U.S.'s transportation sector. This study uses a non-radial SBM-DEA model with an undesirable output (CO_2) to measure the environmental efficiency of the U.S. transportation sector from 2004 to 2012. Using the SBM-DEA model, environmental efficiency (EE), carbon efficiency (CE) and potential carbon reduction (PCR) are estimated for each state, and we measure the size of slack input resources and excess CO_2 emissions as well as the shortfall of desirable output (transportation GDP).

According to the results we draw the following conclusions: (1) most states had an average EE below 0.64 during 2004–2012, meaning that these states had considerable room for improvements in transportation environmental efficiency; (2) among the 50 U.S. states, four states were found to be environmentally efficient (Alaska, Illinois, Nebraska and Vermont), the remaining 46 were inefficient with Alabama, South Carolina, Colorado and Mississippi being the most inefficient with average EE and CE scores below 0.4; (3) there was a large PCR for most of U.S. states, and the average PCR was 7.10 million metric tons; (4) the U.S.'s transportation sector still had great amount of excess in transportation capital and labor, and a potential for more CO_2 emissions reduction.

Striking a balance between adequate transportation provisions and reducing transportation carbon footprint has been a challenging task. The findings of this study provide policy insights as well as an overview of the transportation sector's environmental performance in the U.S. First of all, the slack analysis shows the potential improvement of state-level environmental efficiency performances in the transportation sector through better resource utilization and reduced carbon emissions. Second, the policy should adopt the goal and strategy of encouraging energy conservation to reduce CO_2 emissions in the transportation sector. The DEA benchmarking results of this study show that state policymakers could learn and adopt the best practices in eco-efficient states to enhance transportation environmental efficiency. Finally, the U.S. could improve technological innovation and the current fuel economy standards to produce a more environmentally friendly transportation system.

Although this study provides an overall understanding of the environmental performance of the U.S.'s transportation sector, limitations exist. First of all, individual states' performances were compared with other states in the country, and the

results may be sensitive to the number of inputs and outputs as well as the levels of aggregation in the data. Also, this study uses state-level transportation sector's GDP as the only good output, but state-level transportation output may vary by the geographical location or the composition of the transportation sector within the state. For example, coastal water transportation is not part of the sector in landlocked states in the U.S. continent. Therefore, future research could consider using more disaggregated market data rather than the sector's GDP to try to capture the relative efficiencies of the states. In this paper, we did not consider the technological change of the transportation sector. It is recommended that future studies further examine the dynamic changes of environmental efficiency as well as the technological changes of the sector. Lastly, the environmental efficiency of the sector may also be analyzed using stochastic frontier analysis. A slacks-based directional distance function (Fukuyama and Weber, 2009; Färe and Grosskopf, 2003) may also be used as an alternative to the SBM efficiency.

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